L2.4 Limits at Infinity

Work in your group. Check your answers against the Quarto tonight.

Limits at infinity and horizontal asymptotes

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1. Evaluate. Divide top and bottom by the highest power of x in the **denominator**, and show the division.

$$\lim_{x\to\infty} \frac{4x-3}{x^2+2}$$

2. Evaluate.

$$\lim_{x\to\infty} \frac{6x^2-x}{2x^2+5}$$

The three asymptote rules, derived

3. Identify the horizontal or slant asymptote.

$$f(x) = \frac{x^3 - 2x^2 + 1}{-3x^3 + 2x - 1}$$

4. Identify the horizontal or slant asymptote.

$$g(x) = \frac{x + 1}{x^{99} + 1}$$

5. Identify the horizontal or slant asymptote.

$$h(x) = \frac{3x^6 + x^5}{2x^5}$$

Limits at $-\infty$

6. Evaluate.

$$\lim_{x\to\infty} \frac{\sqrt{9x^2+2}}{2x+5}$$

7. Same function, other direction.

$$\lim_{x\to-\infty} \frac{\sqrt{9x^2+2}}{2x+5}$$

Stop. Your two answers should differ. If they do not, look at what $\frac{1}{x}$ becomes when it goes inside the radical.

8. Sketch a graph of

$$f(x) = \frac{3x^2+4x+5}{\sqrt{16x^4-81}}$$

Find the vertical asymptotes and the horizontal asymptotes first. Say how many horizontal asymptotes there are, and why that is different from the last problem.

Slant asymptotes

9. Find the slant asymptote by polynomial long division.

$$y = \frac{2x^2 + 3x - 1}{x + 1}$$

10. Evaluate.

$$\lim_{x \to \infty} \left(\sqrt{x^2 + 3x} - x \right)$$

Sketching from asymptotes

11. Draw an **even** function which has the lines $y = 1$, $x = -4$ and $x = -1$ among its asymptotes.

12. BOARDS **Group work.** Let

$$P(x) = a_m x^m + \cdots + a_1 x + a_0 \qquad \text{and} \qquad Q(x) = b_n x^n + \cdots + b_1 x + b_0.$$

Your group gets one case. Find $\lim_{x \to \infty} \dfrac{P(x)}{Q(x)}$, **derive it** from the limit at infinity theorem, and be ready to report out.

(a) $m = n$

(b) $m < n$

(c) $m > n$